



Modelling Financial Markets

Seminar 14: Deep Learning and Time-Series Foundation Models — Hands-on



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ ACI: Adaptive Conformal Inference
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CI: Confidence Interval
- ▣ CRPS: Continuous Ranked Probability Score
- ▣ CRSP: Center for Research in Security Prices
- ▣ CW: Clark–West (test for nested models)
- ▣ DM: Diebold–Mariano (test)
- ▣ DQ: Dynamic Quantile (test)
- ▣ ES: Expected Shortfall
- ▣ ETF: Exchange-Traded Fund
- ▣ FHS: Filtered Historical Simulation
- ▣ FRED: Federal Reserve Economic Data
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GPD: Generalised Pareto Distribution
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HS: Historical Simulation
- ▣ LR: Likelihood Ratio (test)
- ▣ LSTM: Long Short-Term Memory (network)
- ▣ MCS: Model Confidence Set
- ▣ MDE: Minimum Detectable Effect

Acronyms used in this material (2/2)

- ▣ **MSE: Mean Squared Error**
- ▣ **OLS: Ordinary Least Squares**
- ▣ **PIT: Probability Integral Transform**
- ▣ **QLIKE: Quasi-LIKElihood (loss function)**
- ▣ **QS: Quantile Score**
- ▣ **RFF: Random Fourier Features**
- ▣ **RV: Realised Variance**
- ▣ **SPA: Superior Predictive Ability (test)**
- ▣ **TS: Time Series (in the name TS-Arena)**
- ▣ **US: United States**
- ▣ **VAR: Vector AutoRegression**

Outline

Introduction

Key Formulas

Part A: Derivations

Part B: Inference on the Forecasts

Part C: Open Question and AI Critique

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive vanishing gradients as VAR impulse responses and the LSTM cell state as an AR(1)
- ▣ Read VaR and ES from a quantile grid, with a GPD tail; compute the exact power of a backtest
- ▣ Test forecasts correctly: Clark–West, SPA, Romano–Wolf, DQ, PIT and quantile scores, conformal recalibration
- ▣ Detect seed effects, stale prices and look-ahead leakage

- ▣ Background: Lecture 14 and Lecture 8 (backtesting); textbook [Franke, Härdle & Hafner \(2019\)](#)

Structure

- ▣ **Part A:** ten derivations (A1–A10)
- ▣ **Part B:** eleven computational exercises with inference (B1–B11)
- ▣ **Part C:** an open question, a seed for the team project (C1), an AI answer to critique (C2), and a project seed from the lecture case study (C3)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Always name the benchmark: zero for returns, log-HAR for volatility, FHS for tail risk
- A model that is "not significantly worse" is not "as good": ask how much power the test had

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (gradients as VAR impulse responses), A3 (mean scaling), A5 (VaR and ES from a grid), A8 (exact power of Kupiec), A10 (QLIKE elicits the mean)
- Part B: B1 (returns vs zero, Clark–West, SPA), B3 (realised volatility vs log-HAR), B5 (VaR 1% and ES 2.5%), B7 (seeds), B9 (PIT, Berkowitz, quantile score)

Proposed: you solve them

- Part A: A2 (model: A1), A4 (model: A3), A6 and A7 (model: A5), A9 (model: A8)
- Part B: B2 (model: B1), B4, B8, B11 (model: B3), B6 (model: B5), B10 (models: B5, B9)
- Part C: C1, open question and project seed; C2, critique of an AI answer (a foundation-model VaR backtest); C3, project seed from the lecture case study (the virtue of complexity in return prediction)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter14_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `torch`, `chronos-forecasting`, `timesfm`, `arch`, `pandas`, `scipy`
- Series: S&P 500, Bitcoin, BET (daily closes); SPY 5-minute bars for realised variance; up to 18 September 2026
- Forecasts: the one-day-ahead forecasts of Lecture 14 (context of 512 days for the foundation models; windows of 500 or 1,000 days for the classical models)
- Returns r_t in %, losses $L_t = -r_t$; $\text{VaR}_\alpha = -q_\alpha$, $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$

Key Formulas (1/2): Networks and Tokens

- LSTM (Hochreiter & Schmidhuber (1997)): $c_t = f_t \odot c_{t-1} + i_t \odot g_t$, $h_t = o_t \odot \tanh(c_t)$; gates $\sigma(Wx_t + Uh_{t-1} + b) \in (0, 1)$
- Parameters of one layer: $4(hd + h^2 + h)$, or $4(hd + h^2 + 2h)$ with two bias vectors (PyTorch)
- Memory along the cell state: $\partial c_T / \partial c_{T-k} = \prod_j f_{T-j}$; half-life $k_{1/2} = \ln 0.5 / \ln f$
- Attention (Vaswani et al. (2017)): $\text{softmax}(QK^\top / \sqrt{d_k}) V$
- Chronos tokens (Ansari et al. (2024)): $s = \frac{1}{\epsilon} \sum |x_t|$, $z_t = x_t / s$, bin index $j = \text{round}((z_t + 15) / \Delta)$, $\Delta = 30/4092$

Key Formulas (2/2): Evaluation

- Out-of-sample $R^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$ (benchmark: zero forecast)
- Diebold–Mariano (Diebold & Mariano (1995)): $DM = \bar{d} / \sqrt{\widehat{LRV} / T}$, $d_t = L_t^A - L_t^B$, Newey–West long-run variance; negative: A better
- QLIKE (Patton (2011)): $RV_t / \hat{h}_t - \log(RV_t / \hat{h}_t) - 1$; HAR (Corsi (2009)): RV on its daily, 5-day and 22-day averages
- Kupiec (Kupiec (1995)): $LR_{uc} = -2[(T - x) \ln(1 - \alpha) + x \ln \alpha - (T - x) \ln(1 - \hat{\pi}) - x \ln \hat{\pi}] \sim \chi_1^2$
- FZ0 (Patton, Ziegel & Chen (2019)): $\frac{1}{\alpha \text{ES}} 1\{L > \text{VaR}\}(L - \text{VaR}) + \frac{\text{VaR}}{\text{ES}} + \log \text{ES} - 1$; MCS (Hansen, Lunde & Nason (2011)) at 90%

Key Formulas (3/3): Inference and Calibration

- Clark–West (Clark & West (2007)), nested benchmark: $\tilde{d}_t = e_{0,t}^2 - [e_{1,t}^2 - (\hat{y}_{0,t} - \hat{y}_{1,t})^2]$, one-sided HAC t
- Many models: SPA (Hansen (2005)), $\max_k \sqrt{T} \bar{d}_k / \hat{\omega}_k$ with a stationary bootstrap; Romano–Wolf stepwise adjusted p (Romano & Wolf (2005))
- DQ (Engle & Manganelli (2004)): regress $I_t - \alpha$ on $(1, I_{t-1}, \dots, I_{t-4}, \text{VaR}_t)$; $\hat{\beta}^\top X^\top X \hat{\beta} / (\alpha(1 - \alpha)) \sim \chi_6^2$
- PIT $u_t = \hat{F}_t(y_t)$ i.i.d. $U(0, 1)$ (Diebold, Gunther & Tay (1998)); Berkowitz LR on $\Phi^{-1}(u_t)$ (Berkowitz (2001)); QS $= \frac{2}{K} \sum_k \text{pinball}_{u_k}$, an equal-weight grid analogue of the CRPS (Gneiting & Raftery (2007))
- Adaptive conformal (Gibbs & Candès (2021)): $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$; $\hat{Q} = +\infty$ when $\alpha_t \leq 0$; long-run breach rate $\rightarrow \alpha$ for any sequence

A1. Vanishing Gradients as VAR Impulse Responses [Solved]

Task

- **Context:** a simple recurrent network $h_t = \tanh(Wh_{t-1} + Ux_t + b)$ with two hidden units, trained by backpropagation through time; no data are needed
 - ▶ Given: $W = \begin{pmatrix} 0.5 & 0.8 \\ 0 & 0.5 \end{pmatrix}$; linearise at the fixed point $h^* = 0$ with $b = 0$ and $x = 0$
- (a) Derive the Jacobian $\partial h_T / \partial h_{T-k}$ with the chain rule and prove that $\|\partial h_T / \partial h_{T-k}\|_2 \leq \|W\|_2^k$
- (b) Show that the linearised network is a VAR(1) in h_t and that the Jacobian equals its impulse response at horizon k
- (c) Compute the spectral radius $\rho(W)$, the spectral norm $\|W\|_2$, the bound $\|W\|_2^k$ and the exact $\|W^k\|_2$ for $k = 5, 20, 50$
- **Report** the numbers of (c) and answer: is the norm bound or the spectral radius the right criterion for vanishing gradients?

A1. Vanishing Gradients as VAR Impulse Responses: solution (1/2)

Solution

- (a) One step: $\partial h_t / \partial h_{t-1} = D_t W$ with $D_t = \text{diag}(1 - h_t^2)$, since $\tanh'(z) = 1 - \tanh^2(z)$
 - ▶ Chain rule over k steps: $\partial h_T / \partial h_{T-k} = \prod_{j=0}^{k-1} D_{T-j} W$
 - ▶ $\|D_t\|_2 = \max_i |1 - h_{t,i}^2| \leq 1$ and $\|AB\|_2 \leq \|A\|_2 \|B\|_2$, hence the product is at most $\|W\|_2^k$
- (b) At $h^* = 0$: $\tanh'(0) = 1$, so $D_t = I$ and $h_t \approx W h_{t-1} + U x_t$
 - ▶ This is a VAR(1) with coefficient matrix W , forced by $U x_t$ (an innovation if x_t is centred white noise)
 - ▶ Its moving-average form $h_T = \sum_k W^k U x_{T-k}$ shows that $\partial h_T / \partial h_{T-k} = W^k$ is the impulse response at horizon k

A1. Vanishing Gradients as VAR Impulse Responses: solution (2/2)

Solution

- (c) W is triangular: its eigenvalues are the diagonal entries, so $\rho(W) = 0.50$; the largest singular value gives $\|W\|_2 = 1.040$
 - ▶ Bound at $k = 20$: $1.040^{20} = 2.20$, which grows with k and says nothing
 - ▶ Exact: $W^k = 0.5^k \begin{pmatrix} 1 & 1.6k \\ 0 & 1 \end{pmatrix}$, so $\|W^5\|_2 = 0.254$, $\|W^{20}\|_2 = 3.1 \times 10^{-5}$, $\|W^{50}\|_2 = 7.1 \times 10^{-14}$
- Interpretation: the norm bound is sufficient, not necessary; the gradient vanishes at the rate $k 0.5^k$ because $\rho(W) < 1$ at this fixed point, where $D = I$
 - ▶ Gelfand's formula $\|W^k\|^{1/k} \rightarrow \rho(W)$: the same criterion as the stability of a VAR; away from $h^* = 0$ it is $\rho(DW)$ that matters, not $\rho(W)$

A2. The LSTM Cell State as an AR(1) [Proposed]

Solution discussed in class

Task

- **Context:** an LSTM cell with a constant forget gate $f = \sigma(b_f)$, a constant input gate i and i.i.d. candidates g_t with mean 0 and variance σ_g^2
 - ▶ Data for (c): S&P 500 daily log returns (GSPC.INDX, data/market) since January 2000; ACF of $|r_t|$: 0.289 (lag 1), 0.132 (lag 50), 0.059 (lag 150)
- (a) Show that $c_t = fc_{t-1} + ig_t$ is a stationary AR(1) for $0 < f < 1$; derive $\text{Var}(c_t)$ and $\text{Corr}(c_t, c_{t-k})$
- (b) Compute the half-life $\ln 0.5 / \ln f$ for $b_f = 1$ and for $f = 0.99$, the bias b_f that gives $f = 0.99$, and f^{250} for $b_f = 5$
- (c) Fit a geometric decay with its own amplitude, $\rho_k = \rho_1 f^{k-1}$ (an AR(1) signal observed with noise): choose f so that $f^{49} = \rho_{50} / \rho_1$; compute b_f , the half-life and the predicted $\rho_{150} = \rho_1 f^{149}$
- **Report** the numbers of (b) and (c); why can no fixed $f < 1$ reproduce long memory, and what does the network need instead?
- Model: A1

A3. Mean Scaling in Chronos: Who Moves the Scale? [Solved]

Task

- **Context:** the original Chronos (Ansari et al. (2024)) divides a context of C returns by $s = \frac{1}{C} \sum_t |x_t|$ and maps $z = x/s$ to 4,093 equally spaced bins in $[-15, 15]$, $\Delta = 30/4092$
 - ▶ Given for (b): a four-day context of returns in %, $x = (0.8, -1.2, 0.4, 2.0)$
- (a) The window moves one day: a new value M enters and x_{out} leaves. Derive s_{new} as a function of s_{old} , M , x_{out} and C , and the factor by which the scaled value of every unchanged day changes
- (b) For the four-day context compute s , the scaled values z and the bin indices $j = \text{round}((z + 15)/\Delta)$
- **Report** the formula of (a) and the table of z and j ; how does the effect of one crash on the scale depend on C ?

A3. Mean Scaling in Chronos: Who Moves the Scale?: solution

Solution

- (a) The sum of absolute values loses $|x_{\text{out}}|$ and gains $|M|$: $\sum_{\text{new}} = \sum_{\text{old}} - |x_{\text{out}}| + |M|$
 - ▶ Dividing by C : $s_{\text{new}} = s_{\text{old}} + (|M| - |x_{\text{out}}|)/C$
 - ▶ Unchanged day x : $z_{\text{new}} = x/s_{\text{new}} = z_{\text{old}} \cdot s_{\text{old}}/s_{\text{new}}$
- (b) $s = (0.8 + 1.2 + 0.4 + 2.0)/4 = 1.10$
 - ▶ $z = x/s = (+0.727; -1.091; +0.364; +1.818)$; $\Delta = 0.00733$
 - ▶ $j = (2145, 1897, 2096, 2294)$: all near the centre of the vocabulary, $j = 2046$ for $z = 0$
- Interpretation: a crash moves s by about $|M|/C$: a short context reacts strongly, a 512-day context hardly at all

A4. A Crash Day in the Context [Proposed]

Solution discussed in class

Task

- ▣ **Context:** the scaling of A3; S&P 500 daily log returns (GSPC.INDX, data/market) in a 512-day context
 - ▶ On 16 March 2020 the return -12.77% enters and a day of 0.51% leaves; before the crash $s_{\text{old}} = 0.741\%$
- ▣ (a) Replace the last value of the A3 context by -9.0 and recompute s , z , j and the factor $s_{\text{old}}/s_{\text{new}}$
- ▣ (b) Apply the formula of A3(a) to the S&P 500 case and compute s_{new} and the factor
- ▣ (c) Compute the scaled value of an ordinary 0.4% day before and after the crash, in both cases
- ▣ **Report** s , the factor and z of the 0.4% day; does a 512-day context make ordinary days look calm after a crash?
- ▣ Model: A3

A5. VaR 1% and ES 2.5% from a Quantile Grid [Solved]

Task

- ▣ **Context:** the last zero-shot Chronos-2 forecast for the S&P 500 (day after 18.09.2026; context of 512 daily returns, GSPC.INDX); grid levels 1%, 5%, 10%, ..., 99%
 - ▶ Given: $q_{0.01} = -2.704\%$, $q_{0.05} = -1.400\%$; convention $\text{VaR}_\alpha = -q_\alpha$, $\text{ES}_\alpha = -\frac{1}{\alpha} \int_0^\alpha q_u du$
- ▣ (a) Compute VaR 1%
- ▣ (b) Interpolate $q_{0.025}$ linearly between $q_{0.01}$ and $q_{0.05}$ and compute VaR 2.5%
- ▣ (c) Compute ES 2.5%, holding $q_u = q_{0.01}$ for $u < 0.01$ and using the linear q_u on $[0.01, 0.025]$
- ▣ **Report** VaR 1%, VaR 2.5%, ES 2.5% and the ratio ES 2.5%/VaR 2.5%; compare this ratio at the same tail level with its value for a zero-mean Normal distribution

A5. VaR 1% and ES 2.5% from a Quantile Grid: solution

Solution

- (a) $\text{VaR } 1\% = -q_{0.01} = 2.704\%$
- (b) $q_{0.025} = q_{0.01} + \frac{0.025-0.01}{0.05-0.01}(q_{0.05} - q_{0.01}) = -2.704 + 0.375(-1.400 + 2.704) = -2.215\%$
 - ▶ $\text{VaR } 2.5\% = -q_{0.025} = 2.215\%$
- (c) Split the integral: $\int_0^{0.01} q_u du = 0.01 q_{0.01}$; $\int_{0.01}^{0.025} q_u du = 0.015 (q_{0.01} + q_{0.025})/2$ (trapezoid)
 - ▶ $\text{ES } 2.5\% = -\frac{1}{0.025} \left[0.01 q_{0.01} + 0.015 \frac{q_{0.01} + q_{0.025}}{2} \right] = 2.557\%$
- Interpretation: $\text{ES } 2.5\%/\text{VaR } 2.5\% = 1.15$, below the value for the zero-mean Normal distribution at the same level, $\varphi(1.96)/(0.025 \cdot 1.96) = 1.19$: the grid cuts the tail below 1% (A6)

A6. Clamping the Grid Tail; a GPD Tail [Proposed]

Solution discussed in class

Task

- **Context:** the grid of A5; S&P 500 daily losses $L = -r$ (GSPC.INDX) over the last 1,000 trading days to 18.09.2026
 - ▶ Peaks over threshold: u = empirical 95% quantile of L ; excesses $L - u$ follow a $\text{GPD}(\xi, \beta)$ with $\xi < 1$ (McNeil & Frey (2000))
- (a) With the quantile function exact on $[0.01, 0.025]$, prove that holding $q_u = q_{0.01}$ for $u < 0.01$ never raises ES 2.5%; then show that the linear interpolation of A5 on $[0.01, 0.025]$ can err in either direction
- (b) Derive $\text{ES}_\alpha = \frac{\text{VaR}_\alpha}{1-\xi} + \frac{\beta-\xi u}{1-\xi}$ from the mean excess function of the GPD
- (c) Fit ξ, β by maximum likelihood; compute $\text{VaR}_\alpha = u + \frac{\beta}{\xi} \left[(n\alpha/N_u)^{-\xi} - 1 \right]$ for $\alpha = 1\%$ and 2.5% , and ES 2.5%
- **Report** u , N_u , $\hat{\xi}$, $\hat{\beta}$, VaR and ES; compare ES/VaR with A5
- **Model:** A5

A7. A “VaR 1%” from Deciles [Proposed]

Solution discussed in class

Task

- **Context:** Chronos-Bolt (small), zero-shot, same day and context as A5; it returns only the deciles $q_{0.1}, \dots, q_{0.9}$
 - ▶ Given: $q_{0.1} = -0.691\%$, $q_{0.5} = 0.071\%$
- (a) State the value returned when VaR 1% is requested from a decile grid that is clamped at its lowest level
- (b) Assume a Normal predictive distribution with median $q_{0.5}$ and $\hat{\sigma} = (q_{0.5} - q_{0.1})/z_{0.9}$; compute VaR 1% $= -(q_{0.5} + \hat{\sigma}z_{0.01})$ and ES 2.5% $= -(q_{0.5} - \hat{\sigma}\varphi(z_{0.025})/0.025)$
- (c) Compare with Chronos-2 (A5) and assess the Normal-extrapolation assumption: which causes of the too-frequent breaches of raw foundation-model quantiles (Lecture 14, Test 3) can it address, and which test would settle it?
- **Report** the three numbers and the comparison
- **Model:** A5

A8. The Exact Power of the Kupiec Test [Solved]

Task

- **Context:** VaR 1% forecasts over the 220 trading days after the release of the weights (3.11.2025–18.09.2026); the number of breaches $x \sim \text{Bin}(T, \pi)$ with $T = 220$
 - ▶ Kupiec (Kupiec (1995)) LR test, two-sided, level 5%: reject if $\text{LR}_{uc} > \chi^2_{1;0.95} = 3.84$
- (a) Compute $\text{LR}_{uc}(x) = -2[(T - x) \ln 0.99 + x \ln 0.01 - (T - x) \ln(1 - x/T) - x \ln(x/T)]$ for $x = 0, 1, 2, \dots$ and find the rejection region
- (b) Compute the exact size $P_{\pi=1\%}(\text{reject})$
- (c) Compute the power $P_{\pi=1.67\%}(\text{reject})$; 1.67% is the breach rate of raw Chronos-2 on the S&P 500
- **Report** the rejection region, size and power; what does a non-rejection over 220 days tell a risk manager?

A8. The Exact Power of the Kupiec Test: solution (1/2)

Solution

- (a) $x = 0$: $LR = -2 \cdot 220 \ln 0.99 = 4.42 > 3.84$: reject (too few breaches)
 - ▶ $x = 5$: $LR = 2.65 < 3.84$; $x = 6$: $LR = 4.51 > 3.84$
 - ▶ Rejection region: $x = 0$ or $x \geq 6$; acceptance region: $1 \leq x \leq 5$
- (b) $\text{Size} = P_{1\%}(x = 0) + P_{1\%}(x \geq 6) = 0.134$, not 0.05: with a discrete statistic the nominal level cannot be hit

A8. The Exact Power of the Kupiec Test: solution (2/2)

Solution

- (c) Power = $P_{1.67\%}(x = 0) + P_{1.67\%}(x \geq 6) = 0.9833^{220} + 0.165 = 0.025 + 0.165 = 0.189$
 - Expected number of breaches: $220 \times 1\% = 2.2$ under H_0 , $220 \times 1.67\% = 3.7$ under the alternative
- Interpretation: four times in five a model with 67% too many breaches passes; over 220 days, “not rejected” is no evidence of correct coverage

A9. How Long Must the Post-Release Window Be? [Proposed]

Solution discussed in class

Task

- ▣ **Context:** the Kupiec test of A8 (two-sided, 5%) for VaR 1%; true breach rate $\pi = 1.67\%$ or 2%
- ▣ (a) Compute the exact power against $\pi = 1.67\%$ for $T = 1,000$ and $T = 2,693$ (the full sample of Lecture 14)
- ▣ (b) Find the smallest T with power $\geq 80\%$ and the T from which it stays above 80%; explain why power is not monotone in T
- ▣ (c) Repeat for $\pi = 2\%$: the power at $T = 220$ and the T for 80% power
- ▣ **Report** the numbers; how many years of post-release data does a clean backtest need?
- ▣ Model: A8

A10. Which Functional Does QLIKE Elicit? [Solved]

Task

- ▣ **Context:** SPY realised variance RV (sum of squared 5-minute returns), one-day-ahead forecasts from 11.10.2022 to 18.09.2026 (979 days); loss QLIKE $L(y, h) = y/h - \log(y/h) - 1$
 - ▶ Chronos-2 on log RV (context 512) returns quantiles q_u ; average ratio of $\exp(q_{0.5})$ to $\int_0^1 \exp(q_u) du$ over the days: 0.8048
- ▣ (a) Minimise $E[L(Y, h) \mid \mathcal{F}]$ over $h > 0$ and show that the minimiser is $E[Y \mid \mathcal{F}]$
- ▣ (b) Assume $\log Y \mid \mathcal{F} \sim N(\mu, s^2)$: derive the median, the mean, their ratio and the expected QLIKE excess of the median forecast
- ▣ (c) From the Chronos-2 ratio infer an effective common s and the QLIKE excess (a day-averaged ratio does not identify each day's s)
- ▣ **Report** the derivations and the two numbers; why is $\exp(\text{median})$ a biased volatility forecast?

A10. Which Functional Does QLIKE Elicit?: solution (1/2)

Solution

- (a) $E[L] = E[Y]/h - E[\log Y] + \log h - 1$; $\partial_h E[L] = -E[Y]/h^2 + 1/h = 0 \Rightarrow h^* = E[Y \mid \mathcal{F}]$
 - Second derivative $2E[Y]/h^3 - 1/h^2 = 1/h^{*2} > 0$ at h^* : a minimum; QLIKE is consistent for the mean (Gneiting (2011))
- (b) Median e^μ , mean $m = e^{\mu+s^2/2}$, ratio $e^{-s^2/2}$
 - With $h = me^{-s^2/2}$: $E[L(h)] - E[L(m)] = e^{s^2/2} - 1 - s^2/2 > 0$

A10. Which Functional Does QLIKE Elicit?: solution (2/2)

Solution

- (c) $s = \sqrt{-2 \ln(\text{ratio})} = \sqrt{-2 \ln 0.8048} = 0.659$; excess $e^{s^2/2} - 1 - s^2/2 = 1/\text{ratio} - 1 + \ln(\text{ratio}) = 0.0254$
 - Observed in Lecture 14: QLIKE 0.259 with the median against 0.231 with the mean, a gap of the same order
- Interpretation: monotone maps preserve quantiles, so $\exp(\text{median of } \log RV)$ is the median of RV, about 20% below its mean; QLIKE rewards the mean

B1. Can Anything Beat Zero for the S&P 500? [Solved]

Task

- ▣ **Context:** S&P 500 daily log returns in % (GSPC.INDX closes, data/market); test period 04.01.2016–18.09.2026 (2 693 days); benchmark: the zero forecast
 - ▶ One-day-ahead forecasts of Lecture 14, using data up to the previous close: historical mean, AR(1) (rolling 1,000 days), LSTM (average of 5 seeds), Chronos-2, Chronos-Bolt and TimesFM-2.5 (median)
- ▣ (a) Compute the out-of-sample $R^2 = 1 - \sum_t (r_t - \hat{r}_t)^2 / \sum_t r_t^2$ of each model with a 95% circular block-bootstrap CI (blocks of 20 days, 1,000 draws)
- ▣ (b) Test each model against zero with the DM test on squared errors (Newey–West, two-sided, 5%) and compute the share of correct signs
- ▣ (c) Apply Clark–West (Clark & West (2007), one-sided) to the nested historical mean and AR(1); apply SPA (Hansen (2005)) and Romano–Wolf (Romano & Wolf (2005)) to the six models (stationary bootstrap, mean block 20 days, 2,000 draws)
- ▣ **Report** one table (R^2 , CI, DM, p , correct signs) and the Clark–West, SPA and Romano–Wolf p -values

B1. Can Anything Beat Zero for the S&P 500?: interpretation questions

Interpretation

- Is a positive R^2 of about 1% statistically meaningful here? Why is 55% correct signs not evidence of skill?
- Does anything beat zero once nesting and the six-model search are accounted for?

B1. Solution (1/3): Procedure

Solution

- Step 1: align forecasts and returns on the 2693 test days; losses $e_{0,t}^2 = r_t^2$ (zero) and $e_{1,t}^2 = (r_t - \hat{r}_t)^2$ (model)
- Step 2: $R^2 = 1 - \sum e_1^2 / \sum e_0^2$; bootstrap: draw blocks of 20 consecutive days with wrap-around, recompute R^2 1,000 times, take the 2.5% and 97.5% percentiles
- Step 3: $d_t = e_{1,t}^2 - e_{0,t}^2$, $DM = \bar{d} / \sqrt{\widehat{LRV} / T}$ with $\lfloor 4(T/100)^{2/9} \rfloor$ Bartlett lags; $p = 2[1 - \Phi(|DM|)]$; reject equal accuracy if $p < 0.05$
- Step 4: Clark–West $\tilde{d}_t = e_{0,t}^2 - (e_{1,t}^2 - \hat{r}_t^2)$, HAC t -statistic, $p = 1 - \Phi(t)$
- Step 5: SPA and Romano–Wolf: studentised gains t_k of $e_0^2 - e_{1,k}^2$; bootstrap the maximum of the recentred t_k ; Romano–Wolf removes rejected models step by step

B1. Solution (2/3): Results

Model	R^2 (%)	95% CI	DM [p]	Correct sign (%)
Historical mean	+0.15	[-0.02; +0.40]	-1.67 [0.095]	54.6
AR(1)	+0.94	[-2.65; +4.40]	-0.47 [0.639]	54.1
LSTM	+0.93	[+0.17; +1.59]	-1.70 [0.090]	53.5
Chronos-2	-0.26	[-0.89; +0.38]	+0.81 [0.418]	54.7
Chronos-Bolt	-1.57	[-2.73; -0.24]	+1.76 [0.078]	54.6
TimesFM-2.5	-3.35	[-4.62; -1.95]	+3.23 [0.001]	53.6

- LSTM: the CI [+0.17; +1.59] excludes zero, but $DM = -1.70$, $p = 0.090 > 0.05$: equal accuracy with zero is not rejected at 5%
- TimesFM-2.5 is significantly worse than zero; AR(1) has a huge CI, driven by 2020
- The S&P 500 rose on 54.6% of days: "always up" already gets that share of signs right

B1. Solution (3/3): Nesting and Many Models

- Zero is nested in the historical mean and in AR(1): under H_0 their estimation noise makes DM undersized
 - ▶ Clark–West: $\tilde{d}_t = r_t^2 - [(r_t - \hat{r}_t)^2 - \hat{r}_t^2]$, one-sided HAC t
 - ▶ With the zero benchmark $\tilde{d}_t = 2r_t\hat{r}_t$: CW tests $E[r_t\hat{r}_t] = 0$, a necessary implication of the martingale-difference null for forecasts built from past data, asymptotically valid under moment conditions and a non-degenerate variance (Clark & West (2006)); DM, SPA and Romano–Wolf test equal MSE, a different hypothesis
 - ▶ One-sided p , DM vs CW: historical mean 0.048 vs 0.007; AR(1) 0.320 vs 0.055
- Six models tested against the same zero forecast: SPA $p = 0.130$ (H_0 : none beats zero)
 - ▶ Romano–Wolf adjusted p : historical mean 0.164, LSTM 0.161
- Reading: CW detects the small drift of the historical mean; once the six-model search is accounted for, nothing beats zero on the S&P 500

B2. Bitcoin and BET [Proposed]

Solution discussed in class

Task

- **Context:** as B1, for Bitcoin (BTC-USD.CC closes, calendar days, 2818 days from 01.01.2019) and the BET (official index, data/market/BET.csv, 2675 days from 04.01.2016)
 - ▶ Stale-price check: the BET-tracking ETF Patria-TVBETETF (adjusted close), joined with the BET on common days before computing returns
- (a) Repeat B1(a)–(b) for both markets
- (b) For the BET LSTM compute the one-sided DM and Clark–West p , Romano–Wolf within the six BET models, SPA for the BET, and the Holm-adjusted p over all 18 model–market tests
- (c) Compute the BET lag-1 autocorrelation, the correlation of the LSTM forecast with r_{t-1} (Lo & MacKinlay (1990)), and R^2 and Clark–West p of the LSTM signal applied to the ETF return of the same day and of the next day
- **Report** the tables of (a) and the numbers of (b)–(c). Model: B1

B2. Bitcoin and BET: interpretation questions

Interpretation

- Which series has predictable daily returns, and which model detects this predictability?
- Does the signal survive multiplicity and stale prices? Relate the answer to Chapter 2

B3. Realised Volatility: Better than log-HAR? [Solved]

Task

- ▣ **Context:** SPY realised variance $RV_t = \sum_{j=1}^{78} r_{t,j}^2$ from 5-minute bars (data/market/intraday/SPY.US_5m.csv), days with at least 70 bars; 979 one-day-ahead forecasts, 11.10.2022–18.09.2026
 - ▶ Models: random walk, HAR and log-HAR (Corsi (2009); rolling 500 days), GARCH-t, LSTM, Chronos-2, Chronos-Bolt, TimesFM-2.5 (grid mean $\int_0^1 \exp(q_u) du$)
- ▣ (a) Compute the mean QLIKE loss (Patton (2011)) of each model
- ▣ (b) Test each model against log-HAR with DM on QLIKE differences (Newey–West, two-sided, 5%)
- ▣ (c) Compute the 90% model confidence set (Hansen, Lunde & Nason (2011); $T_{\max}$ statistic, circular block bootstrap with blocks of 10 days, 1,000 draws); repeat (a)–(b) on the 218 days after 3 November 2025
- ▣ **Report** one table: QLIKE, DM [p], MCS p , QLIKE and DM after the release

Interpretation

- ▣ What does membership of the MCS tell you, and what does it not tell you?
- ▣ Why does the post-release window matter for foundation models only?

B3. Solution (1/2): Procedure

Solution

- Step 1: daily loss $L_{t,m} = \text{RV}_t / \hat{h}_{t,m} - \log(\text{RV}_t / \hat{h}_{t,m}) - 1$ for each model m ; average over the 979 days
- Step 2: $d_t = L_{t,m} - L_{t,\log \text{HAR}}$; DM with Newey–West variance; negative DM: model m better; reject equality if $p < 0.05$
- Step 3: MCS: for the surviving set compute $t_m = (\bar{L}_m - \bar{L}) / \widehat{\text{se}}$, bootstrap $\max_m t_m$; if $p < 0.10$ eliminate the worst model and repeat; the MCS p of a model is the running maximum of these p -values
- Step 4: repeat steps 1–3 on the days from 3.11.2025, when all three sets of weights were public

B3. Solution (2/2): Results

Model	QLIKE	DM vs log-HAR [p]	MCS p	QLIKE (post)	DM (post)
Random walk	0.326	+4.53 [<0.001]	0.00	0.384	+2.81
HAR	0.247	+1.30 [0.195]	0.29	0.270	+1.40
log-HAR	0.234	–	0.95	0.238	–
GARCH-t	0.283	+3.43 [<0.001]	0.03	0.295	+2.35
LSTM	0.284	+1.68 [0.094]	0.29	0.233	-0.52
Chronos-2	0.231	-0.62 [0.537]	1.00	0.242	+0.57
Chronos-Bolt	0.232	-0.18 [0.857]	0.95	0.236	-0.39
TimesFM-2.5	0.233	-0.08 [0.938]	0.95	0.244	+0.60

- Chronos-2 has the lowest QLIKE but is not significantly better than log-HAR (DM -0.62, $p = 0.537$)
- MCS: random walk and GARCH-t are eliminated; the MCS says "not distinguishable", not "equally good"
- After 3.11.2025 the LSTM is best (0.233); no DM test is significant against log-HAR except for the random walk and GARCH-t

B4. Median or Mean? And How Much Context? [Proposed]

Solution discussed in class

Task

- **Context:** the Chronos-2 and TimesFM-2.5 quantile forecasts of log RV of B3 (same 979 days); Chronos-2 also with contexts of 32, 64, 128, 256 and 512 days on the same forecast days
- (a) For both models compute the QLIKE and the ratio mean forecast / mean RV of $\exp(q_{0.5})$ and of $\int_0^1 \exp(q_u) du$ (trapezoid on the grid)
- (b) Test median against mean for Chronos-2 with DM on the QLIKE differences (Newey–West, 5%)
- (c) Compute the Chronos-2 QLIKE for each context length and compare with log-HAR
- **Report** the table of (a), the DM statistic and the QLIKE by context. Model: B3

Interpretation

- Why is the median biased as a variance forecast (A10)?
- What does the context effect say about the memory of volatility?

B5. Backtesting the Tail Forecasts on the S&P 500 [Solved]

Task

- ▣ **Context:** S&P 500 daily log returns (GSPC.INDX), 2 693 days from 04.01.2016; losses $L_t = -r_t$; one-day VaR 1%, VaR 2.5% and ES 2.5%
 - ▶ Six models: HS, GARCH(1,1)- t and FHS (1,000-day window, re-estimated every 20 days), raw Chronos-2 quantiles, FHS with Chronos-2 or TimesFM volatility
- ▣ (a) Count the VaR 1% breaches $I_t = 1\{L_t > \text{VaR}_t\}$; apply the Kupiec test (Kupiec (1995); two-sided, 5%) and the Christoffersen independence test (first-order Markov chain, χ^2_1)
- ▣ (b) Compute the mean FZ0 loss (Patton, Ziegel & Chen (2019)) of (VaR 2.5%, ES 2.5%) and DM tests against FHS (Newey–West, 5%)
- ▣ (c) For FHS and the Chronos-2 hybrid, compute the DQ test (Engle & Manganelli (2004); constant, four lagged hits and VaR_t , χ^2_6)
- ▣ **Report** one table: breaches, rate, Kupiec p , independence p , FZ0, DM against FHS

B5. Backtesting the Tail Forecasts on the S&P 500: interpretation questions

Interpretation

- Why do the raw Chronos-2 quantiles fail while the hybrid passes the coverage test?
- After passing the coverage test, does the hybrid also pass the independence test and match FHS?

B5. Solution (1/2): Procedure

Solution

- ▣ Step 1: breaches $x = \sum_t I_t$, rate $\hat{\pi} = x/T$; LR_{uc} as in A8; reject correct coverage if $p < 0.05$
- ▣ Step 2: independence: count transitions n_{ij} of $I_{t-1} \rightarrow I_t$, $\hat{\pi}_{01} = n_{01}/(n_{00} + n_{01})$, $\hat{\pi}_{11} = n_{11}/(n_{10} + n_{11})$; LR of $\pi_{01} = \pi_{11}$, χ^2_1 (Christoffersen (1998))
- ▣ Step 3: $FZ0 = \frac{\mathbf{1}\{L > v\}(L-v)}{\alpha e} + \frac{v}{e} + \log e - 1$ with $v = \text{VaR } 2.5\%$, $e = \text{ES } 2.5\%$; DM on the differences against FHS
- ▣ Step 4: DQ: regress $I_t - 0.01$ on $(1, I_{t-1}, \dots, I_{t-4}, \text{VaR}_t)$; $DQ = \hat{\beta}^\top X^\top X \hat{\beta} / (0.01 \cdot 0.99)$; reject if $p < 0.05$

B5. Solution (2/2): Results (S&P 500, 2 693 days)

Model	Breaches	Rate (%)	p Kupiec	p indep.	FZ0	DM vs FHS
HS	37	1.37	0.065	0.001	1.309	+2.46
GARCH-t	44	1.63	0.002	0.205	1.026	+1.34
FHS	30	1.11	0.559	0.045	0.999	–
Chronos-2 quantiles	45	1.67	0.001	0.223	1.147	+3.16
FHS + Chronos-2 vol.	25	0.93	0.705	0.021	1.084	+2.65
FHS + TimesFM vol.	27	1.00	0.989	0.029	1.059	+1.89

- Raw Chronos-2: 1.67% breaches, Kupiec $p = 0.001$: its quantiles are too narrow
- Hybrid (Chronos-2): 0.93%, $p = 0.705$: the tail comes from the data, through FHS
- But breaches cluster (independence $p = 0.021$, DQ $p = 0.001$) and FZ0 is significantly worse than FHS (DM +2.65): right on average, too much capital

B6. Bitcoin, and VaR 10% for All Three Models [Proposed]

Solution discussed in class

Task

- **Context:** the six tail models of B5 for Bitcoin (BTC-USD.CC, 2818 calendar days from 01.01.2019); and the 10% quantiles of Chronos-2, Chronos-Bolt and TimesFM-2.5 against HS and FHS in the three markets
- (a) Repeat B5(a)–(b) for Bitcoin
- (b) For VaR 10% compute the breach rates and the Kupiec test with $\alpha = 10\%$ (two-sided, 5%) for the five models in the three markets
- **Report** the table of (a) and a 3×5 table of breach rates with Kupiec p . Model: B5

Interpretation

- Is the narrowness of foundation-model quantiles specific to the 1% level?

B7. The Seed Lottery [Solved]

Task

- **Context:** the return LSTM of Lecture 14: input the last 20 standardised returns, 16 hidden units, MSE loss, Adam, weight decay 10^{-4} , early stopping on the last 20% of up to 4,000 training days; trained once before each test period of B1–B2
 - ▶ Five networks identical except for the random seed (0–4)
- (a) Compute the out-of-sample R^2 against zero of each seed in the three markets
- (b) Compute the R^2 of the average of the five forecasts
- **Report** a 3×6 table of R^2 (%)

Interpretation

- What would you conclude from a single run on Bitcoin?
- Why is averaging seeds a form of regularisation?

B7. Solution (1/2): Procedure

Solution

- Step 1: for each seed s , $R_s^2 = 1 - \sum_t (r_t - \hat{r}_{t,s})^2 / \sum_t r_t^2$
- Step 2: average forecast $\bar{r}_t = \frac{1}{5} \sum_s \hat{r}_{t,s}$ and its R^2
- Step 3: compare the spread across seeds with the size of R^2 ; by convexity of squared error, the ensemble forecast has no greater squared error than the average squared error of its members

B7. Solution (2/2): Results

R^2 (%)	Seed 0	1	2	3	4	Average
S&P 500	+0.55	+0.37	+0.70	+0.54	+1.96	+0.93
Bitcoin	+0.11	+0.03	-0.79	-0.12	+0.24	+0.04
BET	+1.06	+1.01	+1.12	+1.22	+1.12	+1.15

- Bitcoin: seeds range from -0.79% to +0.24%: a single run could "show" skill or its absence
- BET: all five seeds agree; averaging reduces variance without adding bias, like bagging

B8. Leakage Through a Look-Ahead Regressor [Proposed]

Solution discussed in class

Task

- **Context:** SPY log RV as in B3; log-HAR on a rolling 500-day window with regressors $\log RV_{t-1}$ and the 5- and 22-day averages; forecast $\exp(\hat{y}_t + \hat{s}^2/2)$
- (a) Build a wrong version in which the 5- and 22-day averages include the forecast day t
- (b) Compute the QLIKE of the correct and the wrong log-HAR and of Chronos-2 on the same 979 days; test wrong against correct with DM (Newey–West, 5%)
- **Report** the three QLIKE values, the relative gain and the DM test. Model: B3

Interpretation

- How large is the gain from the look-ahead? Could any honest model get it?
- Where can the same error hide in a deep-learning pipeline?

B9. Calibration of the Whole Predictive Distribution [Solved]

Task

- ▣ **Context:** S&P 500 (GSPC.INDX), 2 693 days from 04.01.2016; the raw Chronos-2 grid of 21 quantiles (1%, 5%, 10%, ..., 99%); FHS and GARCH-t (1,000-day window, refit every 20 days) evaluated on the same grid
- ▣ (a) Compute the PIT $u_t = \hat{F}_t(r_t)$ (Diebold, Gunther & Tay (1998)) by linear interpolation between the grid quantiles, censored outside $[q_{1\%}, q_{99\%}]$; apply Pearson's χ^2 to the counts in the 22 grid intervals (21 degrees of freedom)
- ▣ (b) Apply the Berkowitz LR test (Berkowitz (2001)) of $\mu = 0$, $\sigma = 1$ to $z_t = \Phi^{-1}(u_t)$, with a Normal likelihood censored below 1% and above 99% (χ^2_2 , 5%)
- ▣ (c) Compute the quantile score $QS_t = \frac{2}{K} \sum_k (1\{r_t < q_{u_k}\} - u_k)(q_{u_k} - r_t)$ (an equally weighted quantile score; the CRPS would weight each level by its spacing, Gneiting & Raftery (2007)) and DM of Chronos-2 minus FHS, overall, in the centre (10%–90%) and in the tails
- ▣ **Report** the PIT histogram and a table: tail shares, $\hat{\mu}/\hat{\sigma}$, Berkowitz p , χ^2 p , QS

Interpretation

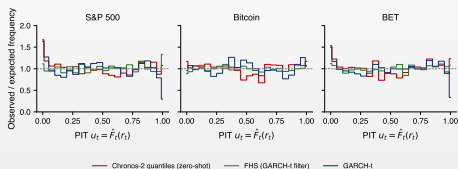
- ▣ Is the failure of the foundation model in the centre or in the tails?
- ▣ Why censor instead of extrapolating beyond the grid?

B9. Solution (1/2): Procedure

Solution

- Step 1: for each model, on its own 21-quantile grid: if $q_{1\%,t} \leq r_t \leq q_{99\%,t}$, u_t = linear interpolation of the levels between the two neighbouring quantiles; otherwise record only “below” or “above”
- Step 2: $\chi^2 = \sum_{j=1}^{22} (O_j - Tp_j)^2 / (Tp_j)$, p_j = width of interval j
- Step 3: $\ell(\mu, \sigma) = \sum_{\text{inside}} \log \frac{1}{\sigma} \varphi\left(\frac{z_t - \mu}{\sigma}\right) + n_L \log \Phi\left(\frac{c_L - \mu}{\sigma}\right) + n_H \log \left[1 - \Phi\left(\frac{c_H - \mu}{\sigma}\right)\right]$, $c_L = \Phi^{-1}(0.01)$, $c_H = \Phi^{-1}(0.99)$
- LR = $2[\ell(\hat{\mu}, \hat{\sigma}) - \ell(0, 1)] \sim \chi^2_2$; reject calibration if $p < 0.05$; $\hat{\sigma} > 1$ means outcomes more dispersed than forecast; the test evaluates the interpolated grid distribution function and only μ, σ , not serial dependence of the PIT
- Step 4: QS on all 21 levels, on the 17 central levels and on the 4 tail levels; DM with Newey–West variance

B9. Solution (2/2): Results



Model	Below $q_1\%$ (%)	Above $q_{99}\%$ (%)	$\hat{\mu} / \hat{\sigma}$	Berkowitz p	$\chi^2 p$	QS
Chronos-2	1.67	1.08	-0.035 / 1.030	0.019	0.004	0.5101
FHS	1.11	1.34	+0.014 / 0.990	0.599	0.747	0.5062
GARCH-t	1.63	0.30	-0.041 / 0.976	0.023	<0.001	0.5063

- Chronos-2: too many returns below its 1% quantile, $\hat{\sigma} > 1$: Berkowitz and χ^2 reject; FHS passes both
- QS, Chronos-2 minus FHS: DM +3.06 (p 0.002); centre +2.51, tails +3.65: worse in both, most in the tails
- GARCH-t: Berkowitz (p 0.023) and χ^2 reject: 1.63% below $q_1\%$ but only 0.30% above $q_{99}\%$, a symmetric t misses the asymmetric tails; censoring avoids inventing a tail the grid does not have

B10. Conformal Recalibration of Chronos-2 [Proposed]

Solution discussed in class

Task

- ▣ **Context:** raw Chronos-2 quantiles of B5, in the three markets (S&P 500 and BET from 2016, Bitcoin from 2019); evaluation on the common days after the first 250 calibration days
- ▣ (a) Correct the 1% and 2.5% quantiles with adaptive conformal inference ([Gibbs & Candès \(2021\)](#)): scores $E_t = \hat{q}_{\alpha,t} - r_t$ on the last 250 days, corrected quantile $\hat{q}_{\alpha,t} - \hat{Q}_{1-\alpha_t}(E)$, update $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$ with $\gamma = 0.001$; shift ES 2.5% by the VaR 2.5% correction (a heuristic: no conformal guarantee for ES)
- ▣ (b) Repeat with $\gamma = 0$ (plain split conformal)
- ▣ (c) Backtest FHS, raw Chronos-2, the Chronos-2 hybrid and both conformal versions: breaches, conditional coverage, DQ ([Engle & Manganelli \(2004\)](#)), FZ0 and DM against FHS (5%)
- ▣ **Report** one table per market. Models: B5, B9

Interpretation

- ▣ Does conformal recalibration make a foundation model a usable VaR model without GARCH?
- ▣ Why can a model with correct coverage still lose the FZ0 comparison?

B11. A Formal Pre/Post-Release Test [Proposed]

Solution discussed in class

Task

- ▣ **Context:** SPY realised variance, Chronos-2 and log-HAR forecasts of B3 (979 days); chosen common post-release cutoff: 3 November 2025, the first Monday after all three sets of weights were public on Hugging Face (Chronos-2: 30.10.2025); 218 days after
- ▣ (a) Compute $d_t = \text{QLIKE}(\text{Chronos-2}) - \text{QLIKE}(\text{log-HAR})$ and its means before and after the release
- ▣ (b) Regress d_t on a constant and $1\{t \geq 3.11.2025\}$ with Newey–West errors ($\lfloor 4(T/100)^{2/9} \rfloor$ lags) and test $\beta_1 = 0$ at 5%
- ▣ (c) Compute the minimum detectable effect of β_1 , $(z_{0.975} + z_{0.8}) \text{se}_{HAC}(\hat{\beta}_1)$, and its limit when the post-release window grows without bound (n_{pre} fixed); can the test ever reach 80% power for a gap of the pre-release size?
- ▣ **Report** the means, $\hat{\beta}_1$, t , p , MDE. Model: B3

Interpretation

- ▣ Does a non-significant $\hat{\beta}_1$ show that Chronos-2 did not memorise the test period?
- ▣ What design would give the test power?

C1. Can a Zero-Shot Foundation Model Replace GARCH for the BET? [Proposed]

Solution discussed in class

Task

- ▣ **Context:** a risk manager proposes Chronos-2 for the daily VaR 1% and ES 2.5% of a BET position; data: official BET index (data/market/BET.csv), daily log returns, 2 675 days from 04.01.2016 to 18.09.2026; post-release window from 3.11.2025 (217 days)
- ▣ (a) Backtest VaR 1% of the six models of B5: breaches, Kupiec, Christoffersen independence and DQ (5%)
- ▣ (b) Compare (VaR 2.5%, ES 2.5%) with FZ0, DM against FHS and the 90% MCS
- ▣ (c) Classify ten non-overlapping 250-day blocks into Basel zones (green ≤ 4 , yellow 5–9, red ≥ 10 breaches); repeat (a) on the post-release window
- ▣ (d) Pre-register: submit your VaR forecasts for the next 20 trading days before they occur and backtest them afterwards
- ▣ **Report** a one-page recommendation with the tables of (a)–(c)

C1. Can a Zero-Shot Foundation Model Replace GARCH for the BET?: interpretation questions

Interpretation

- Accept, reject, or accept as a challenger model?
- Project extension: fine-tune a foundation model on Romanian stocks and re-run the test

C2: Critique an AI Answer [Proposed]

- A student asks an AI assistant: “Backtest zero-shot Chronos-2 one-day-ahead VaR 1% for the S&P 500 on the last 1500 days, with a 512-day context”
- The assistant answers with this code (spx: daily closes, imports omitted):

```
pipe = BaseChronosPipeline.from_pretrained('amazon/chronos-2')
r = 100 * np.log(spx).diff().dropna().to_numpy()      # daily returns, %
test = np.arange(len(r) - 1500, len(r))
q01 = np.empty(len(test))
for i, t in enumerate(test):                          # forecast of r[t]
    ctx = torch.tensor(r[t - 511:t + 1], dtype=torch.float32) # 512 days
    q, _ = pipe.predict_quantiles([ctx[None, :]], prediction_length=1,
                                quantile_levels=[0.01])
    q01[i] = q[0][0, 0, 0].item()
x = int((r[test] < q01).sum())                          # breaches
print(f"VaR1%={q01.mean():.2f}%,{breaches}{x}")
```

- “Output: VaR 1% = −2.41%, breaches = 5, i.e. 0.33% against 15 expected: the model is conservative, so it passes the Kupiec test easily”

C2: Critique an AI Answer, Task [Proposed]

Solution discussed in class

Task

- ▣ **Context:** the code and conclusion of the previous slide; S&P 500 daily closes, last 1 500 days, zero-shot Chronos-2 with a 512-day context; the answer contains **three** errors, in the code and in the conclusion
- ▣ (a) For each error, say what is wrong and why
- ▣ (b) Say how you would detect it: an index check, the VaR convention of the lecture, the test statistic
- ▣ (c) Give the corrected line or conclusion, and compute the Kupiec statistic and its p -value for 5 breaches in 1 500 days
- ▣ Log the three errors as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, the correct version

Interpretation

- ▣ Which element of $\mathbf{x}$ should be the last one in the context when forecasting $\mathbf{x}[t]$?
- ▣ Is “too few breaches” good news for a risk manager? For a regulator?
- ▣ The weights of Chronos-2 were published in autumn 2025: do the last 1 500 days provide a clean test for it?

C3. The Virtue of Complexity on Course Data [Proposed]

Solution discussed in class

Task

- **Context:** the case study of Lecture 14 (Kelly, Malamud & Zhou (2024)) times the US stock market with ridge regressions on up to 12 000 random Fourier features of the 15 predictors of Welch & Goyal (2008); you apply its method to seven of its predictors on the course data, monthly, from 1993 (an application, not an exact replication)
 - ▶ **Model:** B1 for the evaluation workflow only; here out-of-sample $R^2 = 1 - \text{Var}(\text{error})/\text{Var}(R)$ (footnote 40 of the paper), not the zero-forecast R^2 of B1
- (a) Build the monthly data: SPY total returns from month-end adjusted closes (data/market) minus the previous month's TB3MS/1200 (FRED; an approximation, since TB3MS is an annualised discount rate)
 - ▶ Predictors dated t , in decimals: $\text{tbl} = \text{TB3MS}/100$, $\text{lt} = \text{GS10}/100$, $\text{tms} = \text{lt} - \text{tbl}$, $\text{dfy} = (\text{BAA} - \text{AAA})/100$, $\text{infl} =$ monthly change of CPIAUCSL lagged one month (revised data, not real time), $\text{svar} =$ sum of squared daily S&P 500 log returns in month t , and the excess return of month t as predictor of month $t + 1$
- (b) Standardise as in Section V.A of the paper: returns by their trailing 12-month uncentred standard deviation, predictors by an expanding-window standard deviation, after 36 months of warm-up
- (c) Generate 12 000 random Fourier features with $\gamma = 2$ (eq. 20); for P on a grid of 24 values from 2 to 12 000 (each sine-cosine pair counts as two features) and $\log_{10} z = -3, \dots, 3$, estimate ridge without intercept on rolling 12-month windows, with features scaled by their training-window standard deviation, and form the timing return $\hat{\beta}' S_t R_{t+1}$; average all statistics over 1 000 draws of the features
- (d) Estimate the linear kitchen sink on your seven predictors with $z = 0^+$ and $z = 10^3$ on the same windows
- **Report** the curves of $\|\hat{\beta}\|^2$, out-of-sample R^2 and Sharpe ratio against $c = P/T$ (as in Figures 7–8), and a table in the format of Table I for the two linear models and RFF with $c = 1\,000$, $z = 10^3$: R^2 , annualised Sharpe ratio with t , information ratio against SPY and against the linear model with t , maximum loss, skewness

C3. The Virtue of Complexity on Course Data: interpretation questions

Interpretation

- Your kitchen sink has $c = 7/12$, against $15/12$ in the paper: is it still near the interpolation boundary, and what does the ridgeless curve of Figure 7 predict for its R^2 ?
- Can the Sharpe ratio be positive while $R^2 < 0$? Explain with the split of the forecast error into a correlation part and a scale part (footnote 5 of the paper)
- With about 30 years of test data, what is the power of a two-sided 5% test for an information ratio of 0.3? Use the expected $t \approx \text{ratio} \times \sqrt{\text{years}}$ and a Normal approximation
- Project extension: replace the random features by the internal representations (embeddings) of a pretrained foundation model and repeat (c)

Take-Home Assignment (Team Project)

- ▣ Choose one asset not covered in class (a BVB stock, a crypto-asset, an exchange rate or a commodity)
- ▣ Forecast its daily volatility or VaR 1% with one foundation model, zero-shot, and with a GARCH benchmark
 - ▶ Rolling origin, and the context window must end the day before the forecast
 - ▶ Report the period after the model's release date separately
- ▣ Compare the models with a DM test and, for VaR, with the coverage and independence tests
- ▣ Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- ▣ Deliver a Quantlet folder (code, Metainfo, charts) and three slides: numbers, interpretation, limitations

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